Machine Learning vs. Deep Learning: Real-World Applications

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Introduction

Machine learning (ML) and deep learning (DL) are both parts of artificial intelligence, but they do not approach every problem in the same way. Traditional machine learning often works well when information can be organized into meaningful features, while deep learning is useful when a system needs to learn complicated patterns from large amounts of raw data. Customer churn prediction and medical image recognition show how the type of data and the goal of a project can guide the choice between these approaches.

This paper compares the two approaches through real-world applications: customer churn prediction at Dialog Axiata using traditional machine learning and diabetic retinopathy detection using deep learning. By examining why each approach is suitable for its application, the comparison demonstrates how data characteristics and business or practical objectives influence the selection of an AI technique. The comparison also considers the practical strengths and limitations of each approach.

1. Traditional Machine Learning: Customer Churn Prediction

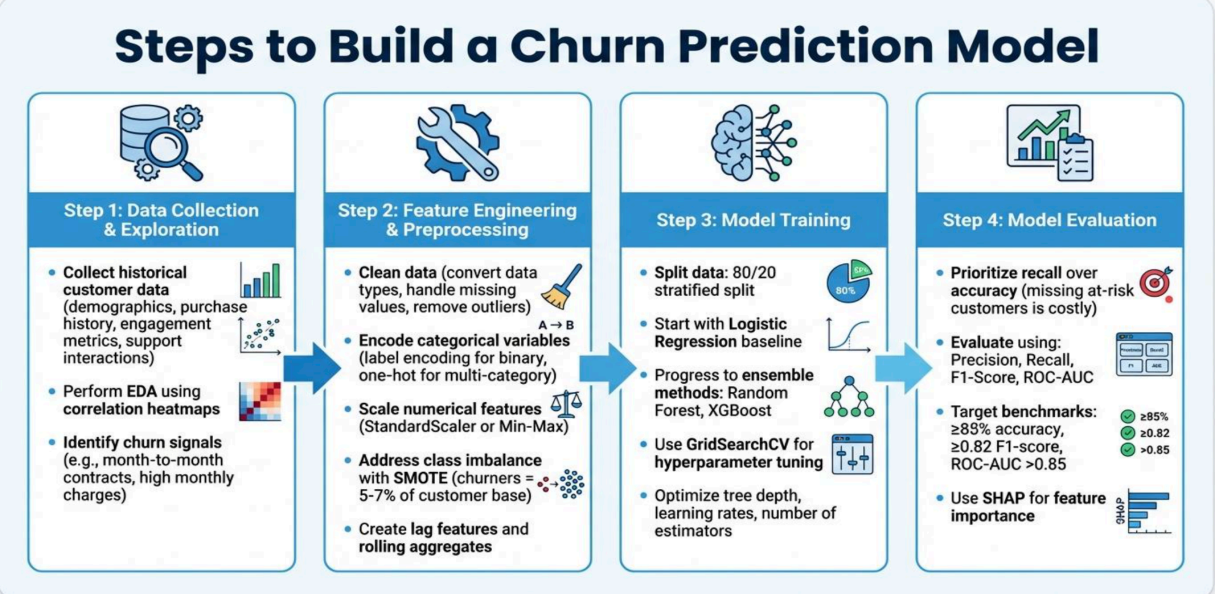
Real-World Application

A real-world example of customer churn prediction comes from Dialog Axiata, a telecommunications company in Sri Lanka. The company developed a Home Broadband Churn Prediction Model to identify customers who might stop using its service. According to Palraj and Ramanayake (2024), the model uses customer information that includes demographic details, network usage, network outages, payment information, complaints, and other service-related features. Its base model uses CatBoost, a gradient-boosted decision-tree algorithm, alongside an

ensemble model. The system produces churn-risk predictions in advance so the company can contact at-risk customers and plan retention campaigns.

Figure 1

Traditional machine learning workflow for customer churn prediction.



This example matches the customer churn prediction topic in the lesson. The lesson describes using customer characteristics such as age, contract length, and monthly charges with a traditional classification algorithm. Dialog Axiata's implementation uses a different traditional ML algorithm, but it addresses the same problem: using past and current customer data to predict which customers may leave.

Why Traditional Machine Learning Is Suitable

Traditional machine learning is suitable for this problem because much of the information used for churn prediction can be represented as structured features, such as usage levels, payment history, service complaints, and network performance. Algorithms such as decision trees and gradient-boosted trees can learn relationships between these features and previous churn

outcomes. They can also rank or explain which factors contribute to a prediction, which helps a business understand what may be causing customers to leave.

The problem is also a classification task: the company wants to identify customers who are likely to leave so it can take action. A traditional ML model can provide risk scores without requiring the company to process every record through a large, multi-layer neural network. For this type of tabular business data, tree-based models can offer a practical balance of predictive performance, training cost, and interpretability. Dialog Axiata's use of CatBoost and an ensemble model provides a concrete example of this approach in practice.

Why Deep Learning Is Not the First Choice for This Example

Deep learning could also be used to predict churn, so it would be inaccurate to say that it cannot solve this problem. However, it may not be the most practical first choice when the main inputs are structured customer records. A deep neural network can require more data, computing resources, tuning, and monitoring. It may also be harder to explain why a particular customer received a high churn-risk score.

A deep learning approach could become more useful if the company wanted to combine its tabular records with large amounts of unstructured information, such as call recordings, customer messages, or voice interactions. For the documented Dialog Axiata use case, traditional ML is a suitable choice because the company can learn from structured customer and network features and use the predictions to guide retention actions.

2. Deep Learning: Medical Image Recognition

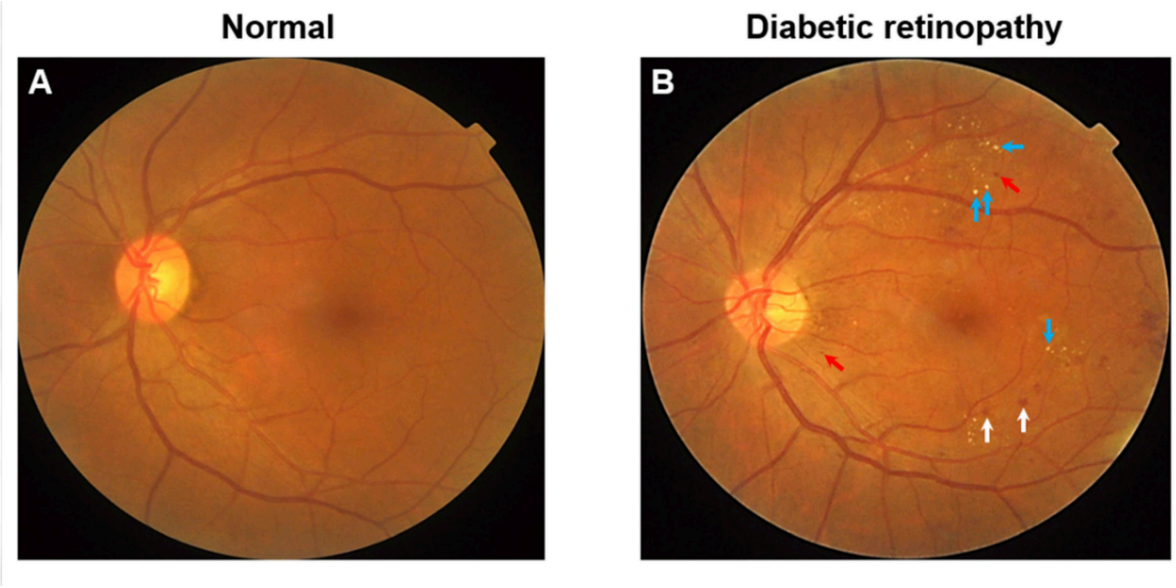
Real-World Application

A real-world example of deep learning in image recognition is the detection of diabetic retinopathy from retinal photographs. Diabetic retinopathy is an eye condition associated with

diabetes that can damage vision. Gulshan et al. (2016) developed and validated a deep learning algorithm that analyzed retinal fundus photographs to identify signs of diabetic retinopathy. Their study showed that a system could learn visual patterns in these images and help identify cases that may need further examination.

Figure 2

Example of retinal fundus imaging used in diabetic retinopathy screening.



The use of this technology was also studied in clinical settings. Beede et al. (2020) examined a deep learning system used in eleven clinics in Thailand and studied how it fit into nurses' screening workflows and patients' experiences. Their work is important because it shows that applying an image-recognition model in healthcare involves more than model accuracy; image quality, clinic processes, and the people using the system also matter. The system is intended to support screening and clinical decisions, not replace a comprehensive eye examination.

Why Deep Learning Is Suitable

Deep learning is suitable for medical image recognition because retinal photographs contain many visual details, including small changes in color, texture, blood vessels, and lesions. It would be difficult to describe every useful visual feature manually before training a model. A convolutional neural network (CNN), the type of architecture highlighted in the lesson for image recognition, can learn visual features from image pixels through multiple layers. Earlier layers can respond to basic patterns such as edges and textures, while deeper layers can combine those patterns into more complex visual information.

This automatic feature learning is valuable for a task such as identifying signs of diabetic retinopathy, where relevant patterns can be subtle and may appear in different parts of an image. With a sufficiently large and representative set of labeled images, a deep learning model can learn which visual patterns are associated with the condition. The approach is therefore well suited to analyzing complex images at scale, provided that the model is carefully validated and used within appropriate clinical procedures.

Why Traditional Machine Learning Is Less Suitable

Traditional machine learning can classify medical images, but it usually needs useful features to be identified or extracted before the algorithm makes a prediction. For retinal photographs, a developer might need to design measurements for color, texture, vessel changes, or suspected lesions. This can be difficult because important signs may be subtle and vary between patients, cameras, and image conditions.

A traditional ML model could still be effective if the images were small in number or if specialists had already identified a reliable set of features. However, when the goal is to learn directly from large collections of complex retinal images, deep learning has an advantage

because it can learn the features and their relationships from the image data. For this reason, deep learning is the stronger fit for this image-recognition problem, while traditional ML is more dependent on the quality of the manually designed features.

Comparison and Conclusion

These examples show that the choice between machine learning and deep learning should depend on the problem and the data, not on the assumption that one method is always better. Dialog Axiata's customer churn problem relies heavily on structured customer and network information, making a traditional tree-based ML model a practical choice. In contrast, diabetic retinopathy detection requires a system to recognize complex visual patterns in retinal photographs, which makes deep learning a better fit.

The comparison also shows why the alternative approach is not always the most practical one. Deep learning may add unnecessary complexity to a structured churn-prediction task, while traditional ML may depend too heavily on manually selected features for complex medical images. Both approaches can be useful, but selecting the one that fits the data, resources, and intended use leads to a more appropriate solution.

References

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